[1]:
```
%pip install tensorflow numpy
```

```
Requirement already satisfied: tensorflow in c:\users\anjali\anaconda3\lib\site-packages (2.21.0)
Requirement already satisfied: numpy in c:\users\anjali\anaconda3\lib\site-packages (1.26.4)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (2.5.
1)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.
6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
(25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in c:\users\anjali\anaconda3\lib\site-packages (fr
om tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
(0.2.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (18.
1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (3.
4.0)
Requirement already satisfied: packaging in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (24.1)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflo
w) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
```

[2]:
```
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
```

[3]:
```
sentence = "the cat sits on the mat"

words = sentence.split()
vocab = sorted(set(words))

word_to_id = {word: i for i, word in enumerate(vocab)}
id_to_word = {i: word for word, i in word_to_id.items()}

print("Words:", words)
print("Vocabulary:", word_to_id)
```

```
Words: ['the', 'cat', 'sits', 'on', 'the', 'mat']
Vocabulary: {'cat': 0, 'mat': 1, 'on': 2, 'sits': 3, 'the': 4}
```

[4]:
```
X = []
y = []

for i in range(1, len(words) - 1):
    context = [
        word_to_id[words[i-1]],
        word_to_id[words[i+1]]
    ]
```

Code JupyterLab Python [conda env:base

```
    target = word_to_id[words[i]]

    X.append(context)
    y.append(target)

X = np.array(X)
y = np.array(y)

print("Input (X):")
print(X)

print("Target (y):")
print(y)
```

```
Input (X):
[[4 3]
 [0 2]
 [3 4]
 [2 1]]
Target (y):
[0 3 2 4]
```

[5]:
```
model = keras.Sequential([
    layers.Input(shape=(2,)),
    layers.Embedding(len(vocab), 10),
    layers.GlobalAveragePooling1D(),
    layers.Dense(len(vocab), activation="softmax")
])
```

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 2, 10)	50
global_average_pooling1d (GlobalAveragePooling1D)	(None, 10)	0
dense (Dense)	(None, 5)	55

Total params: 105 (420.00 B)

Trainable params: 105 (420.00 B)

Non-trainable params: 0 (0.00 B)

```
model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
```

```
model.fit(X, y, epochs=200, verbose=0)
```

```
print("Model training completed!")
```

```
Model training completed!
```

```
context_words = ["cat", "on"]

context_ids = np.array([[
    word_to_id[context_words[0]],
    word_to_id[context_words[1]]
]])

prediction = model.predict(context_ids, verbose=0)

predicted_id = np.argmax(prediction)
predicted_word = id_to_word[predicted_id]

print("Context words:", context_words)
print("Predicted word:", predicted_word)
print("Probability:", round(float(prediction[0][predicted_id]), 3))
```

```
Context words: ['cat', 'on']
Predicted word: sits
Probability: 0.408
```